THE ZERO PARADOX

ZP-L: Incomputability Convergence

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All theorems §I–§VII proved sorry-free in Lean 4. Axiom footprint: [propext, Classical.choice, Quot.sound] throughout this document's own Lean file (Gentzen.lean). Across the ZP layers the footprints are not uniform — ZP-K Section IV tabulates them.

ZP-L establishes four results connecting the formal axioms of the ZP framework to standard results in ordinal theory and computability. First, the axiom footprints of the layers tabulated in Section I are surveyed, and they are not uniform; ZP-K Section IV tabulates the measured footprints. Second, Rogers' fixed-point theorem for a total computable transformation (Mathlib fixed_point; inter-derivable with Kleene's second recursion theorem, fixed_point_2) is formalized as a wrapper, formalizing the computational fixed-point structure. Third, the ordinal ε_0 is fully characterized as the first fixed point of $\alpha \mapsto \omega^\alpha$ and the limit of the tower ω, ω^ω, ω^{ω^ω}, ... Fourth, cnfToZp2 maps ordinals below ε_0 into $\mathbb{Z}_2$ by recursion on their Cantor normal form. From tower stage 1 on, the 2-adic valuation of the image climbs with ordinal height toward ∞ while the 2-adic norm falls toward 0; the images converge to 0, the point the seed stage 0 itself maps to (snap_arc_z2_loop, ZeroParadox/Ordinal/CnfBridge.lean). That 0 is least in $\mathbb{Z}_2$'s norm preorder and greatest in its valuation preorder (z2_two_views, ZeroParadox/Valuation/Scale.lean), the two readings just given.

The central result (§VII) is the canonical snap map: $\phi\,\alpha =$ if $\alpha < \varepsilon_0$ then c_0 else c_1 simultaneously satisfies all five conditions — monotone, tower-aligned, fixed-point-respecting, snapping at ε_0, and ε_0 minimal. All conditions are verified without free hypotheses for this witness. Proved sorry-free, axiom footprint [propext, Classical.choice, Quot.sound] throughout ZeroParadox/Ordinal/Gentzen.lean, measured.

Section I: Axiom Footprint Convergence

Non-constructibility appears across the layers listed below. The axiom footprints are not uniform: ZP-K Section IV tabulates the measured footprints, and the ZPJ/K row below names a witness on each side — bot_self_mem, which measures no axioms, and botCode, which carries them.

Layer	Formal Language	Result cited, or status
ZPB	Topology	C3: no continuous path $\bot \to x \neq \bot$
ZPC	Information Theory	L-INF: surprisal unbounded in depth n, every value finite (l_inf); at Q_2's floor 0 the complexity cxQ2 is $\top$ (infinitude_forces_infinite_complexity, instance q2InfinitudeFloor). That c_0 corresponds to that floor is a reading
ZPJ/K	Set Theory + Computation	bot_self_mem (AFA); botCode (Kleene)

Layer	Formal Language	Result cited, or status
ZPI	Algorithmic IT	K uncomputable. ZP-I § II.B refutes the valuation-complexity bridge (unbounded 2-adic valuation of S_n implies $K(S_n \mid n) / \lvert S_n \rvert \to 1$) with $S_n = 2^n$, the powers of 2 that §IV's tower images take; the valuation half is in Lean, the complexity half is argued outside it

Remark: K is not computed in Lean

Kolmogorov complexity K is not computed in Lean in this framework. The AFA/Kleene route needs no K: da1_paths_unified is a conjunction of two facts, that ⊥ of the lattice is its Quine atom (bot_is_quine_atom) and that the code botCode is a computational quine (the KleeneStructure field botCode_is_quine).

Axiom footprint evidence — the following ZP-K theorems all carry [propext, Classical.choice, Quot.sound]:

t_comp (T-COMP three-way equivalence; the computational face is an assumption, not a clause)

da1_paths_unified

isComputationalQuine_undecidable

infinite_quine_family

Gentzen.lean carries that footprint throughout, measured — and it is not inherited from the theorems above. Mathlib's ordinal theory carries it on its own: Ordinal.nfp and Ordinal.epsilon_zero_eq_nfp each measure [propext, Classical.choice, Quot.sound] with no computability module in scope. ZP-K Section IV tabulates the measured footprints for the computability side.

Section II: Rogers Fixed-Point Stability

Rogers' fixed-point theorem (Mathlib fixed_point; inter-derivable with Kleene's second recursion theorem, fixed_point_2) states that any total computable transformation of a code has a behavioral fixed point: a code c such that running f(c) and running c produce the same partial function. For any computable transformation, at least one fixed-point code exists.

Theorem: roger_fixed_point_stability (Gentzen.lean §II)

For any computable f : Code → Code,

∃ c : Code, eval (f c) = eval c

Proof: wrapper around roger_fixed_point_exists.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Note: this is an existential result. The specific code c is not constructively produced. ZP-K Section IV tabulates the measured footprints.

Section III: The Ordinal ε_0

The ordinal ε_0 is the smallest fixed point of the map $\alpha \mapsto \omega^\alpha$. It is the supremum of the tower $0, 1, \omega, \omega^\omega, \omega^{\omega^\omega}, \ldots$ — each stage is strictly below ε_0, and ε_0 is not reached by any finite iteration. This section is entirely within Lean scope via Mathlib's ordinal machinery.

Definitions (Gentzen.lean §III)
epsilonZero : Ordinal := Ordinal.epsilon 0 (= nfp (ω^·) 0 = veblen 1 0)
fundamentalSeq : ℕ → Ordinal := fun n ↦ (α ↦ ω^α)^[n] 0
Explicit stages:
fundamentalSeq 0 = 0
fundamentalSeq 1 = 1
fundamentalSeq 2 = ω
fundamentalSeq 3 = ω^ω
fundamentalSeq (n+1) = ω^(fundamentalSeq n)

Theorem: epsilonZero_fixedPoint
$\omega^{\varepsilon_0} = \varepsilon_0$
Proof: Ordinal.omega0_opow_epsilon 0 (Mathlib).
Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_eq_nfp
$\varepsilon_0 = \text{nfp}\ (\omega^\cdot)\ 0$
Proof: Ordinal.epsilon_zero_eq_nfp (Mathlib).
Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_eq_iSup
$\varepsilon_0 = \bigsqcup n : \mathbb{N}$, fundamentalSeq n
Proof: iSup_iterate_eq_nfp (Mathlib).
Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_tower_lt
$\forall\ n : \mathbb{N}$, fundamentalSeq $n < \varepsilon_0$

Theorem: epsilonZero_tower_lt

Every finite stage of the tower is strictly below ε_0.

Proof: Ordinal.iterate_omega0_opow_lt_epsilon_zero (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_le_fixedPoint

$\forall\, b : \text{Ordinal},\ \omega^{\wedge} b = b \to \varepsilon_0 \le b$

ε_0 is the least fixed point of $\omega^{\wedge}\cdot$ above 0.

Proof: Ordinal.epsilon_zero_le_of_omega0_opow_le (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: fundamentalSeq_strictMono

$\forall\, n : \mathbb{N},\ \text{fundamentalSeq}\ n < \text{fundamentalSeq}\ (n + 1)$

The tower is strictly monotone.

Proof: by induction; successor step uses isNormal_opow.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Remark R-L.1: Proof-Theoretic Alignment

Two results pin ε_0 as the proof-theoretic ordinal of Peano Arithmetic — the limit of how far PA's proofs of transfinite induction reach — and they run in opposite directions. Gentzen reports, crediting Hilbert-Bernays, that transfinite induction up to any ordinal strictly below ε_0 is provable in PA. And transfinite induction up to ε_0 itself is not provable in PA, which Gentzen proved directly in 1943. Neither half alone locates the boundary: a ceiling with no witnesses beneath it pins nothing, and witnesses with no ceiling pin nothing either. A third result stands behind them — transfinite induction up to ε_0 suffices to prove PA consistent (1936) — and it is also the indirect route to the unprovability half, via Gödel's theorem together with that 1936 result. This is not claimed or proved here.

ZP-L shares its ε_0 with proof theory: it is Mathlib's Ordinal.epsilon 0 (epsilonZero, ZeroParadox/Ordinal/Gentzen.lean), and the ordinals below it are written in Cantor normal form, the notation ordinal analysis also uses (Gentzen 1943, §1). What is ZP-L's own is the snap: a monotone map sending the tower's stages to c_0 fires nowhere below ε_0 (snap_threshold_is_epsilon_zero), and its firing at ε_0 is the hypothesis $h\varepsilon_0$ (ZeroParadox/Ordinal/Incompleteness.lean § II). That the snap's least possible firing point is the ordinal of Gentzen's consistency proof is offered as a structural observation, not as a link between the two results.

Remark R-L.2: Surreal Numbers

Every ordinal is a surreal number (Conway, 1976), so ε_0 lives in the surreal number field No. No satisfies the axioms of a real-closed field, and by Tarski's completeness theorem for the theory of real-closed fields, every first-order sentence in the language of ordered fields that holds in $\mathbb{R}$ also holds in No, and vice versa.

Remark R-L.2: Surreal Numbers

ZP-F (The Counterexamples) proves that the Binary Snap — the transition from $\bot$ to the ordinal threshold ε_0, the first fixed point of $\omega^{\cdot}$ above $\bot$ of Ordinal (leastness: epsilonZero_le_fixedPoint, this section) — cannot occur in any linearly ordered field. This result applies directly to No considered as a linearly ordered field. Note the scope: what is derived is the transition's SHAPE and its impossibility in an ordered field.

The surreals therefore contain ε_0 as an ordinal while simultaneously satisfying the density condition that blocks the Binary Snap in their ordered field structure. Both structures coexist in No: ε_0 is present as an ordinal, and the field density that blocks the snap is present in the field structure. The two results apply to different structural aspects of No.

Remark R-L.3: Hyperreals and Łoś's Theorem

The hyperreals $^*\mathbb{R} = \mathbb{R}^{\mathbb{n}}/U$ (ultrafilter U on $\mathbb{N}$) satisfy a transfer result: by Łoś's theorem, every first-order sentence true in $\mathbb{R}$ holds in $^*\mathbb{R}$. Łoś's theorem applies to the full first-order theory, so any first-order property of $\mathbb{R}$ — including field density — transfers.

Field density transfers: between any two hyperreals there is another. The density condition that blocks the Binary Snap in $\mathbb{R}$ (proved in ZP-F) therefore holds in $^*\mathbb{R}$ as well.

The non-standard naturals $^*\mathbb{N}$ contain infinite elements, but every infinite $H \in {}^*\mathbb{N}$ has a predecessor $H-1$ in $^*\mathbb{N}$. ε_0 is a limit ordinal: it has no predecessor and is the supremum of the tower below it. $^*\mathbb{N}$ cannot host the Binary Snap not only because of field density, but because its infinite elements are successor-like rather than limit-like. The snap requires genuine limit ordinal structure.

The monad of 0 in $^*\mathbb{R}$ — the external set $\{\varepsilon : |\varepsilon| < 1/n \text{ for all standard } n\}$ — is external to the ultrapower. Between any two distinct elements of the monad there is another (their arithmetic mean in $^*\mathbb{R}$), so no discrete jump at 0 is possible in $^*\mathbb{R}$.

Field density is an internal property of the ordered field structure of $^*\mathbb{R}$ and transfers by Łoś's theorem. Limit ordinal structure is set-theoretic and is not preserved by the ultrapower construction: $^*\mathbb{N}$ contains only successor-like infinite elements, not limit-like ones. The two results — transfer of density, failure of limit-ordinal structure — come from different theorems.

Section IV: Cantor Normal Form Bridge

Every ordinal below ε_0 has a unique Cantor normal form — a finite sum $\omega^{e_1}\cdot a_1 + \omega^{e_2}\cdot a_2 + \ldots$ with strictly decreasing exponents $e_1 > e_2 > \ldots$ and each coefficient a nonzero natural number (a nonzero ordinal strictly below ω). In Lean: NONote (Mathlib.SetTheory.Ordinal.Notation). The map cnfToZp2 sends each such ordinal to $\mathbb{Z}_2$ by structural recursion on the Cantor normal form. From tower stage 1 on, the 2-adic valuation of the image tracks ordinal height (tower_orders_agree, ZeroParadox/Ordinal/CnfBridge.lean).

Definition: cnfToZp2 (Gentzen.lean §IV)

cnfToZp2 : NONote → $\mathbb{Z}_2$

Base: cnfToZp2(0) = 0

Definition: cnfToZp2 (Gentzen.lean §IV)

Recursive: cnfToZp2(ω^e · n + a) = 2^(v_2(cnfToZp2(e)) + 1) · n + cnfToZp2(a) [n : ℕ⁺]

where v_2 denotes Lean's PadicInt.valuation, which sets $v_2(0) = 0$; the recursion uses that value.

Valuation of the n-th tower stage:

cnfToZp2(towerNONote 0) = 0 (Lean: PadicInt.valuation 0 = 0; the standard 2-adic valuation of 0 is +∞)

cnfToZp2(towerNONote 1) = 2 (valuation 1)

cnfToZp2(towerNONote 2) = 4 (valuation 2)

cnfToZp2(towerNONote n) = 2^n (valuation n; n ≥ 1)

Theorem: cnfToZp2_tower_valuation

∀ n : ℕ, (cnfToZp2 (towerNONote n)).valuation = n

The 2-adic valuation of the image of the n-th tower stage equals n (at n = 0, by Lean's convention PadicInt.valuation 0 = 0).

Proof: by induction; successor step uses PadicInt.valuation_pow.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: cnfToZp2_valuation_unbounded

∀ k : ℕ, ∃ α : NONote, k ≤ (cnfToZp2 α).valuation

The 2-adic valuation is unbounded across ordinals below ε_0.

Proof: towerNONote k witnesses the bound.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: tower_converges_to_zero

Filter.Tendsto (fun n ↦ cnfToZp2 (towerNONote n)) Filter.atTop (nhds 0)

The images of the tower stages converge to 0 (nhds 0) in $\mathbb{Z}_2$; the theorem names no order.

Proof: each stage's image cnfToZp2(towerNONote (n+1)) = 2^(n+1) in $\mathbb{Z}_2$, so its 2-adic norm is ‖2‖^(n+1) = (1/2)^(n+1) → 0. Uses Metric.tendsto_atTop and exists_pow_lt_of_lt_one.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Section V: Ordinal Tower Limit and Snap Threshold

This section connects the ordinal tower to the ZPE machine-phase snap. The cofinality theorem shows that the fundamental sequence approaches ε_0 from below. The lower-bound theorem shows that any monotone tower-aligned map must assign c_0 to all pre-ε_0 ordinals. A canonical witness snapping exactly at ε_0 is exhibited.

What this does NOT claim (§V)

That ε_0 is the proof-theoretic ordinal of PA (not claimed — see Remark R-L.1).

Any statement about formal provability in PA.

That ε_0 is the UNIQUE minimal snap boundary: snap_threshold_is_epsilon_zero shows no ordinal below ε_0 works (for maps satisfying the stated hypotheses), but does not rule out maps that snap at some ordinal strictly above ε_0.

That the snap threshold result applies to all maps Ordinal → MachinePhase regardless of the monotonicity and tower-alignment hypotheses.

Theorem: fundamentalSeq_cofinal

$\forall\, \alpha : \text{Ordinal}, \alpha < \varepsilon_0 \to \exists\, n : \mathbb{N}, \alpha < \text{fundamentalSeq } n$

The fundamental sequence is cofinal in ε_0: for any ordinal below ε_0, some tower stage exceeds it.

Proof: $\varepsilon_0 = \text{nfp } (\omega^{\wedge}\cdot)\ 0$ (epsilonZero_eq_nfp), and lt_nfp_iff gives $a < \text{nfp } f\ b \leftrightarrow \exists\, n, a < f^{[n]}\ b$.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: snap_threshold_is_epsilon_zero

For any $\phi : \text{Ordinal} \to \text{MachinePhase}$ satisfying:

(a) hmono: $\forall\, \alpha \le \beta$, join $(\phi\ \alpha)\ (\phi\ \beta) = \phi\ \beta$ (order-non-decreasing in the ZPSemilattice sense: $\phi\ \alpha \le \phi\ \beta$)

(b) h0: $\forall\, n : \mathbb{N}, \phi\ (\text{fundamentalSeq } n) = c_0$

we have: $\forall\, \alpha < \varepsilon_0, \phi\ \alpha = c_0$

This is a lower bound: no ordinal below ε_0 is a snap point for maps satisfying (a) and (b). "Minimal" not "unique."

Proof: cofinality gives a stage above α; monotonicity chains $\phi\ \alpha \le \phi(\text{stage}) = c_0$; $c_0 = \bot$ forces $\phi\ \alpha = c_0$.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: c1_epsilon_zero_identification

$\exists\, \phi : \text{Ordinal} \to \text{MachinePhase},$

$(\forall\, n : \mathbb{N}, \phi\ (\text{fundamentalSeq } n) = c_0) \wedge \phi\ \varepsilon_0 = c_1$

Witness: $\phi\ \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$.

One specific map sends all tower stages to c_0 and ε_0 to c_1.

Theorem: c1_epsilon_zero_identification

The witness map is order-non-decreasing in ZP-A's join order (snap_map_mono), and no map Ordinal → MachinePhase is compatible with the CNF → $\mathbb{Z}_2$ map along the tower: for every g : MachinePhase → $\mathbb{Z}_2$ the square fails, because MachinePhase has two values and the images of stages 1, 2, 3 have three distinct 2-adic valuations (the example after c1_epsilon_zero_identification in ZeroParadox/Ordinal/Gentzen.lean); through snapEmbed it fails at every stage n ≥ 1, because snapEmbed takes only the values 1 and 0, and stage n ≥ 1 maps to 2^n, which is neither (ZeroParadox/Ordinal/Incompleteness.lean § II).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Section VI: Kleene-Ordinal Fixed-Point Bridge

The ordinal fixed-point structure ($\varepsilon_0 = \text{nfp}\ (\omega^{\wedge}\cdot)\ 0$, $\omega^{\wedge}\varepsilon_0 = \varepsilon_0$) and the computational fixed-point structure (Kleene's recursion theorem, roger_fixed_point_stability) both carry Classical.choice in the proofs recorded here — parallel structure, not a proved isomorphism. The hypothesis hfp encodes that ordinal fixed points of $\omega^{\wedge}\cdot$ map to the snap state c_1. Under this hypothesis plus monotonicity and tower alignment, ε_0 is the minimal snap threshold.

Theorem: snap_exactly_at_epsilon_zero

For any ϕ : Ordinal → MachinePhase satisfying:

(a) hmono: order-non-decreasing (join ($\phi\ \alpha$) ($\phi\ \beta$) = $\phi\ \beta$ for $\alpha \le \beta$)

(b) h0: every tower stage maps to c_0

(c) hfp: every fixed point of $\omega^{\wedge}\cdot$ maps to c_1

we have: $\phi\ \varepsilon_0 = c_1$ AND $\forall\ \alpha$, $\phi\ \alpha = c_1 \to \varepsilon_0 \le \alpha$

ε_0 is the minimal snap threshold: ϕ assigns c_1 first at ε_0.

Note: "minimal" not "unique." A map satisfying these hypotheses could also assign c_1 to ordinals above ε_0; what is ruled out is any snap strictly before ε_0.

Proof: upper bound from hfp + epsilonZero_fixedPoint; lower bound from snap_threshold_is_epsilon_zero.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: kleene_ordinal_snap_bridge

$\exists\ \phi$: Ordinal → MachinePhase,

$(\forall\ n : \mathbb{N},\ \phi\ (\text{fundamentalSeq}\ n) = c_0) \wedge$

$(\forall\ \alpha,\ \omega^{\wedge}\alpha = \alpha \to \phi\ \alpha = c_1) \wedge$

$\phi\ \varepsilon_0 = c_1 \wedge$

Theorem: kleene_ordinal_snap_bridge

$\forall\, \alpha,\ \phi\ \alpha = c_1 \to \varepsilon_0 \le \alpha$

Witness: $\phi\ \alpha =$ if $\alpha < \varepsilon_0$ then c_0 else c_1.

Note: the theorem name refers to the informal §VI conceptual parallel between Kleene recursion fixed points and ordinal fixed points of $\omega^{\cdot}$. The theorem itself is purely ordinal — no Code or eval appears.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Section VII: Canonical Snap Map — Full Closure

The canonical threshold map $\phi\ \alpha =$ if $\alpha < \varepsilon_0$ then c_0 else c_1 satisfies all three conditions of snap_exactly_at_epsilon_zero (hmono, h0, hfp). For this specific map, the snap identification is unconditional: all five conditions are simultaneously verified without free hypotheses.

Theorem: snap_map_mono

$\forall\, \alpha\ \beta : \text{Ordinal},\ \alpha \le \beta \to$

join (if $\alpha < \varepsilon_0$ then c_0 else c_1)

(if $\beta < \varepsilon_0$ then c_0 else c_1) =

if $\beta < \varepsilon_0$ then c_0 else c_1

The canonical map is order-non-decreasing.

Proof: four cases (α/β vs ε_0). The case $\alpha \ge \varepsilon_0$ with $\beta < \varepsilon_0$ contradicts $\alpha \le \beta$. Each live case closes by rfl from the MachinePhase join definition.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilon_zero_snap_canonical

$\exists\, \phi : \text{Ordinal} \to \text{MachinePhase},$

$(\forall\, \alpha\ \beta,\ \alpha \le \beta \to \text{join}\ (\phi\ \alpha)\ (\phi\ \beta) = \phi\ \beta) \wedge$ [hmono]

$(\forall\, n : \mathbb{N},\ \phi\ (\text{fundamentalSeq}\ n) = c_0) \wedge$ [h0]

$(\forall\, \alpha,\ \omega^{\wedge}\alpha = \alpha \to \phi\ \alpha = c_1) \wedge$ [hfp]

$\phi\ \varepsilon_0 = c_1 \wedge$ [snap]

$\forall\, \alpha,\ \phi\ \alpha = c_1 \to \varepsilon_0 \le \alpha$ [minimality]

All five conditions verified for the explicit witness $\phi\ \alpha =$ if $\alpha < \varepsilon_0$ then c_0 else c_1, with no free hypotheses.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: snap_zp2_correspondence

Four independent facts about the same tower sequence:

(i) $\forall\, n : \mathbb{N}$, fundamentalSeq n $< \varepsilon_0$

(ii) $\forall\, n : \mathbb{N}$, ϕ (fundamentalSeq n) $= c_0$ where ϕ is the canonical map

(iii) Filter.Tendsto (fun n ↦ cnfToZp2 (towerNONote n)) Filter.atTop (nhds 0)

(iv) $\phi\ \varepsilon_0 = c_1$

The same tower sequence witnesses both the ordinal approach to ε_0 and the 2-adic approach to 0. This is a co-witness, not an identity (cnf_bridge_type_boundary, ZeroParadox/Ordinal/CnfBridge.lean): ε_0 and the $\mathbb{Z}_2$ zero live in different types with no coercion between them, so an identity between them fails to elaborate.

Proof: ⟨epsilonZero_tower_lt, fun n ↦ if_pos …, tower_converges_to_zero, if_neg (lt_irrefl ε_0)⟩

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

The Bridge: What Is Formal and What Is Assumed

What the Bridge Supplies

c_1 is a MachinePhase state and ε_0 an ordinal, so an identity between them is not a well-formed statement; the formal link runs through maps: a canonical map Ordinal → MachinePhase and snapEmbed : MachinePhase → $\mathbb{Z}_2$.

What is proved: a canonical map Ordinal → MachinePhase assigns c_1 exactly at ε_0 and nowhere earlier.

Two of the three pieces a type bridge from the ordinals through MachinePhase to $\mathbb{Z}_2$ was expected to need are built, in ZPM (ZeroParadox/Ordinal/Incompleteness.lean §I–§II), and the third is not supplied by them:

(1) snapEmbed : MachinePhase → $\mathbb{Z}_2$ exists, sending $c_0 \mapsto 1$ and $c_1 \mapsto 0$ — the snap state to 2-adic zero, the pre-snap state to a unit (snapEmbed_c0, snapEmbed_c1, both by rfl)

(2) snapEmbed_mul_morphism proves snapEmbed (join a b) = snapEmbed a × snapEmbed b, carrying join to multiplication because c_1 absorbs joins as 0 absorbs products. That is an absorbing-element morphism, and it does not send bottom to bottom in the Scale.lean chart ($\bot = 0$): there the bottom role is filled by 0 (a ZPSemilattice on $\mathbb{Z}_2$ with $\bot = 0$ is built in ZeroParadox/Valuation/Scale.lean § V), and snapEmbed sends c_0 to 1 (snapEmbed_c0); in the multiplicative reading snapEmbed $c_0 = 1$ is the identity ($1 \cdot x = x$), as c_0 is the identity of join, and that reading is not a ZPSemilattice, since multiplication on $\mathbb{Z}_2$ is not idempotent ($2 \cdot 2 \neq 2$)

(3) not supplied — for a map ϕ sending the tower's stages to c_0 (h0), continuity of snapEmbed ∘ ϕ along the tower would force $\phi\ \varepsilon_0 = c_0$, the opposite (the examples after snap_unconditional, ZeroParadox/Ordinal/Incompleteness.lean § II), while the cnfToZp2 images of the stages do tend to snapEmbed $c_1 = 0$; the placement enters as $h\varepsilon_0$.

What the Bridge Supplies

So hfp is derived rather than assumed: hfp_from_epsilon_zero obtains it from monotonicity together with $h\varepsilon_0 : \phi\ \varepsilon_0 = c_1$, and snap_unconditional uses it in place of the free hypothesis of snap_exactly_at_epsilon_zero. $h\varepsilon_0$ ($\phi\ \varepsilon_0 = c_1$; in the $\mathbb{Z}_2$ chart, snapEmbed ($\phi\ \varepsilon_0$) = 0) is the snap's occurrence at ε_0, taken as a hypothesis: of the infinitely many admissible firing points that monotonicity and tower alignment (every tower stage sent to c_0) leave open, $h\varepsilon_0$ selects the least, and so fixes ϕ uniquely (ZeroParadox/Ordinal/Incompleteness.lean § II). Whether Classical.choice is forced by the metric collapse is a separate open question (ZeroParadox/Ordinal/SyntacticCollapse.md). The canonical witness (epsilon_zero_snap_canonical) satisfies all five conditions without any such type bridge.

Theorem Summary

Theorem	Section	Status
roger_fixed_point_stability	§II	Proved ✓
epsilonZero_fixedPoint	§III	Proved ✓
epsilonZero_eq_nfp	§III	Proved ✓
epsilonZero_eq_iSup	§III	Proved ✓
epsilonZero_tower_lt	§III	Proved ✓
epsilonZero_le_fixedPoint	§III	Proved ✓
fundamentalSeq_strictMono	§III	Proved ✓
tower_stage_zero / _one / _two	§III	Proved ✓
towerNONote_repr	§IV	Proved ✓
cnfToZp2_tower_valuation	§IV	Proved ✓
cnfToZp2_valuation_unbounded	§IV	Proved ✓
fundamentalSeq_zp2_converges	§IV	Proved ✓
tower_converges_to_zero	§IV	Proved ✓
zpe_snap_ordinal_correspondence	§V	Proved ✓
epsilonZero_tower_bound	§V	Proved ✓
c1_epsilon_zero_identification	§V	Proved ✓
fundamentalSeq_cofinal	§V	Proved ✓
snap_threshold_is_epsilon_zero	§V	Proved ✓
snap_exactly_at_epsilon_zero	§VI	Proved ✓
kleene_ordinal_snap_bridge	§VI	Proved ✓

Theorem	Section	Status
snap_map_mono	§VII	Proved ✓
epsilon_zero_snap_canonical	§VII	Proved ✓
snap_zp2_correspondence	§VII	Proved ✓

Axiom Purity

Every theorem in the Theorem Summary above carries axiom footprint: [propext, Classical.choice, Quot.sound]. Section I's table and ZP-K Section IV record where the ZP layers differ — Section I's own ZPJ/K row names bot_self_mem, which measures no axioms.

These are axioms of Lean's core library, reached here through Mathlib's ordinal theory, p-adic analysis and computability, and in one place through ZP-K's own Classical.choose (machinePhaseKleene, below). They are not ZP-L commitments.

The footprints across the ZP layers are not uniform; ZP-K Section IV tabulates the measurements. In the computability layer, Rogers' fixed-point theorem as Mathlib states it carries Classical.choice in its statement, through Mathlib's numbering of program codes (the Denumerable Code instance; ZeroParadox/Computability/Kleene.md § VIII). ZP-K's machinePhaseKleene also picks botCode with Classical.choose, which is what makes that instance noncomputable.

The footprints listed here are copies; to re-measure one, run #print axioms on the named declaration. ZeroParadox/Ordinal/Gentzen.lean, this document's own Lean file, prints those of the results it holds.